

Overview of the “Project PLATEAU”

The Initiative of Digital Twin in Japan

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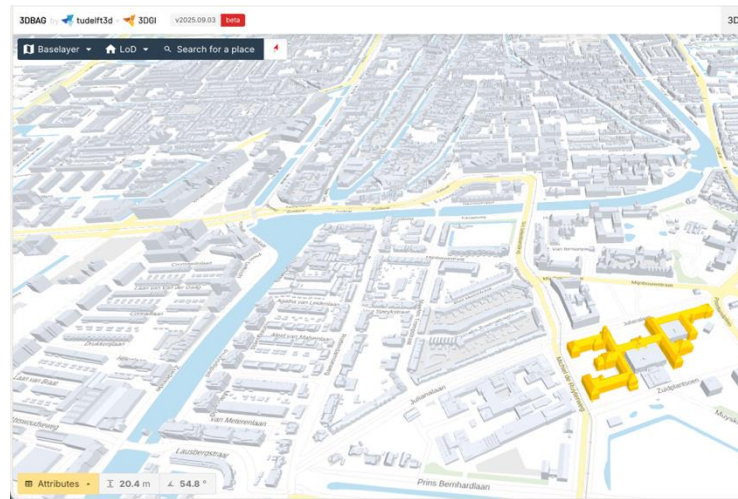
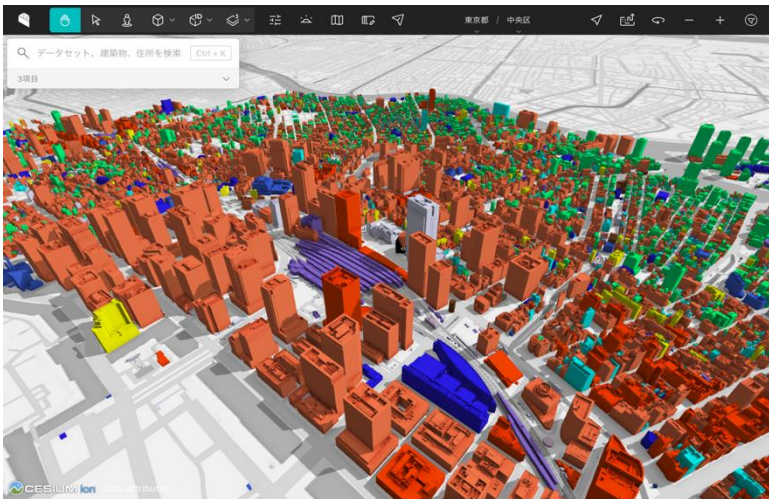
Expert member, “Project PLATEAU” Consortium Advisory Board

Advisory Committee on Urban Policy for the Social Implementation of Digital Twins (MLIT)

Sabbatical Research at TU Delft

(3D geoinformation research group: 2026.04-2027.03)

- **Developing Sustainable Open 3D Spatial Data Infrastructures (SDI) Framework**
- Research Questions:
 1. Technical sustainability of nationwide 3D geospatial infrastructures
 2. Governance and ecosystem design for urban digital twins
 3. Capacity building and education for the utilization of 3D geospatial data.
- Comparative assessment study between Japan and the Netherlands



Agenda

1. What is Project PLATEAU
2. PLATEAU Standards
3. Key Achievements: Use Case Developments and Open Community
4. Future Directions: PLATEAU Vision 2026

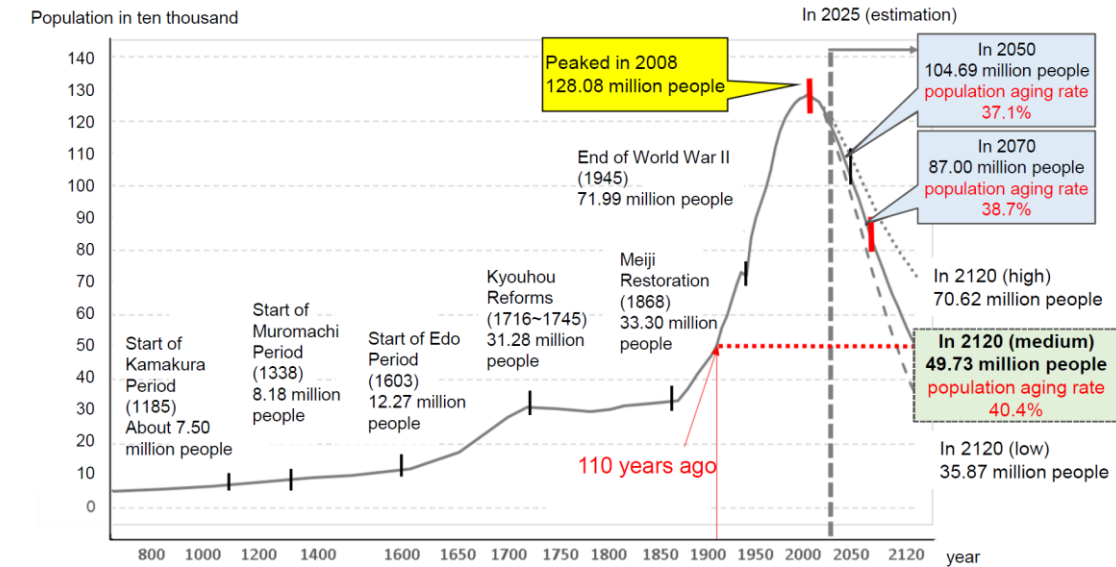
Key Message:

From a national digital twin initiative to a sustainable urban digital public good ecosystem.

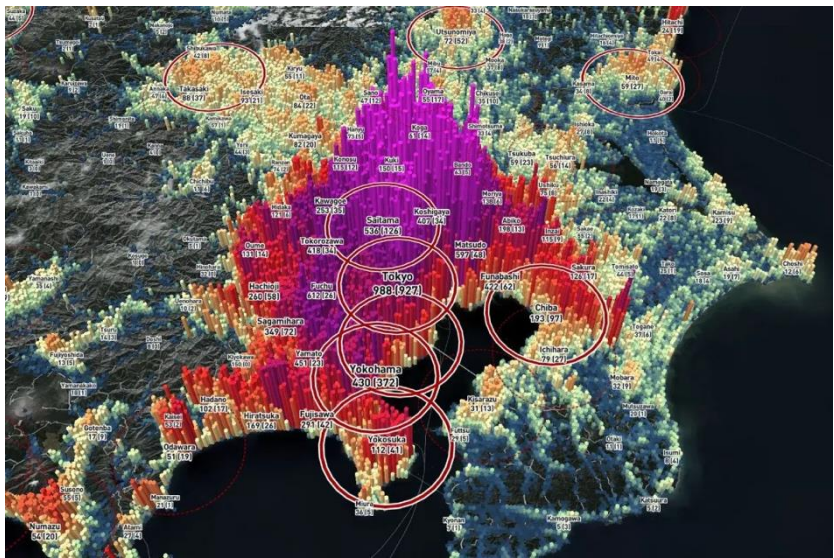
Policy background: What Japan is facing with...

Long-term trends in Japan's population

Japan's population peaked in 2008, and is likely to **return to the level of about 110 years ago over the next 100 years**. In addition, **the ageing rate** is expected to **remain at around 40%**.



Population of the Tokyo metropolitan area: 33 million



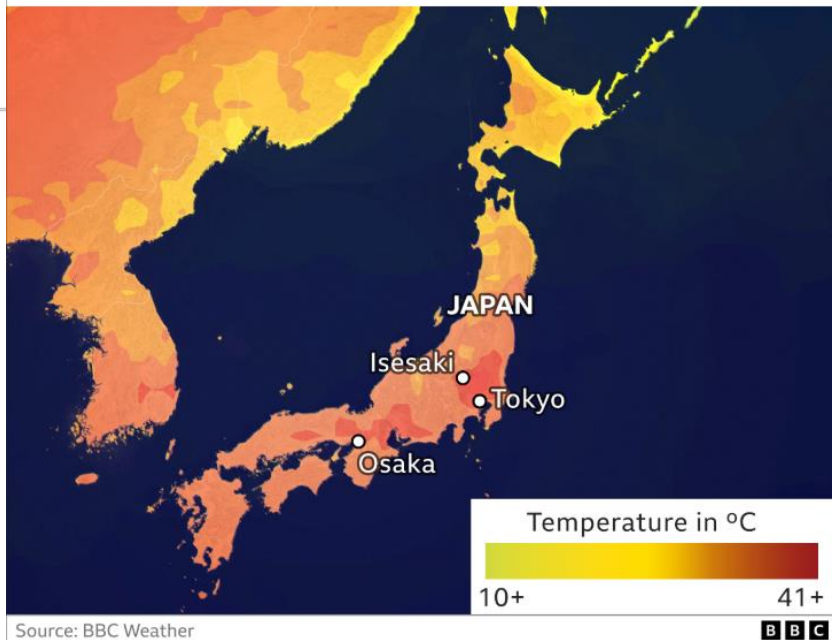
Increasing heavy rainfalls

The frequency of short duration heavy rainfall (over 50mm/hour) has increased by about 1.4 times in 30 years.



Japan is dealing with an intense heatwave

Maximum daily temperature forecast for 29 June



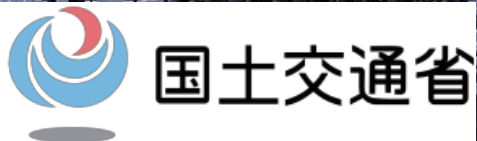
Project PLATEAU

Since 2020

PLATEAU is an **MLIT*-led initiative** to develop and share 3D city models as urban digital twin data, openly providing datasets, tools, and related resources under **CC BY 4.0 and ODbL** licenses to support broad reuse and community collaboration.

Project Mission:

***We create a new value in a society
and solve local issues through the digital twin.***



* MLIT: Ministry of Land, Infrastructure, Transport and Tourism in Japan

Project PLATEAU's Scope

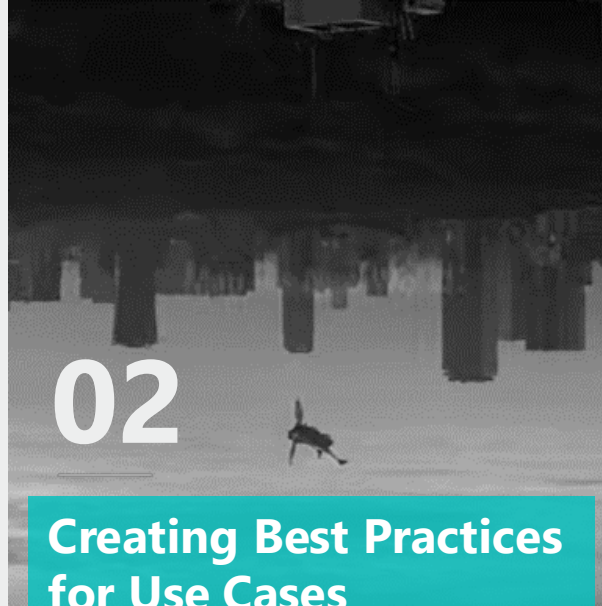
Project PLATEAU is a collaborative initiative led by MLIT, aiming to implement urban digital twins. The project involves partnerships with local governments, private companies, and tech communities.



01

Expanding the coverage of data

To promote data development and updating of 3D city models, we establish standard data models and develop autonomous data development process.



02

Creating Best Practices for Use Cases

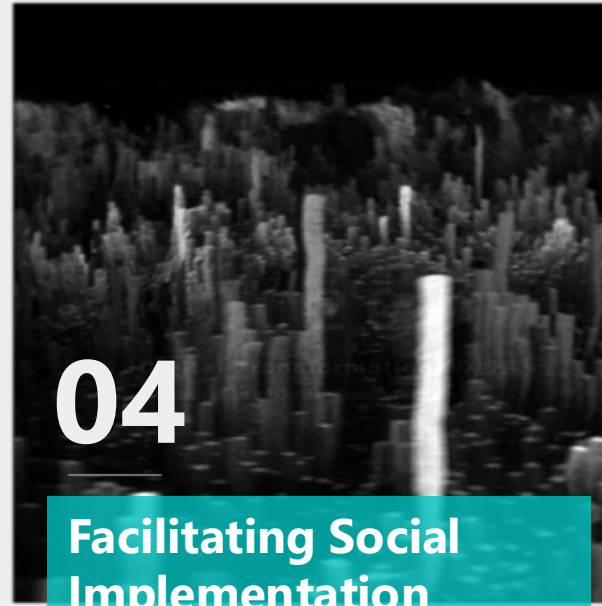
To solve social issues and create new value, We develop solutions that utilize 3D City Models.



03

Fostering Open Innovation

To Foster open innovation, we provide 3D city models as open data and enable communities to leverage them.



04

Facilitating Social Implementation

We provide support for local government to take the initiative in data development and use case creation for 3D city models.

FY2020 - FY2025 Project PLATEAU Summary

304 (→329)
/ 1,741 Cities

30,000km²
/ 378,000km²

Total area



27.6 million*
LOD1

Number of
buildings



about
51.4%
(/1.2billion)

Population
coverage rate

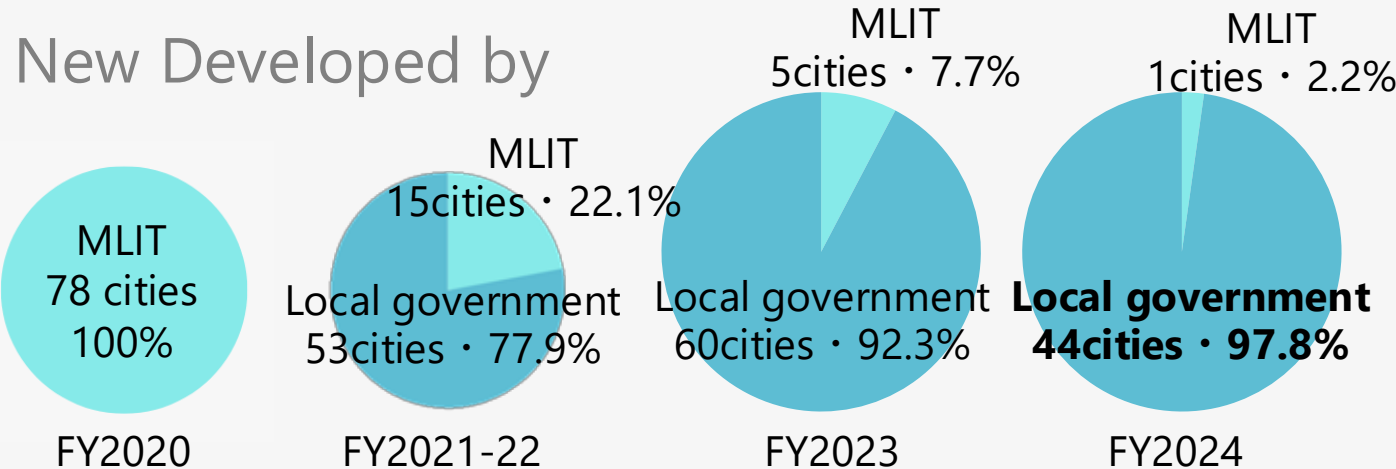


Avg. Budget
€ 11-14 million
/ fiscal year

Development
+ Utilization



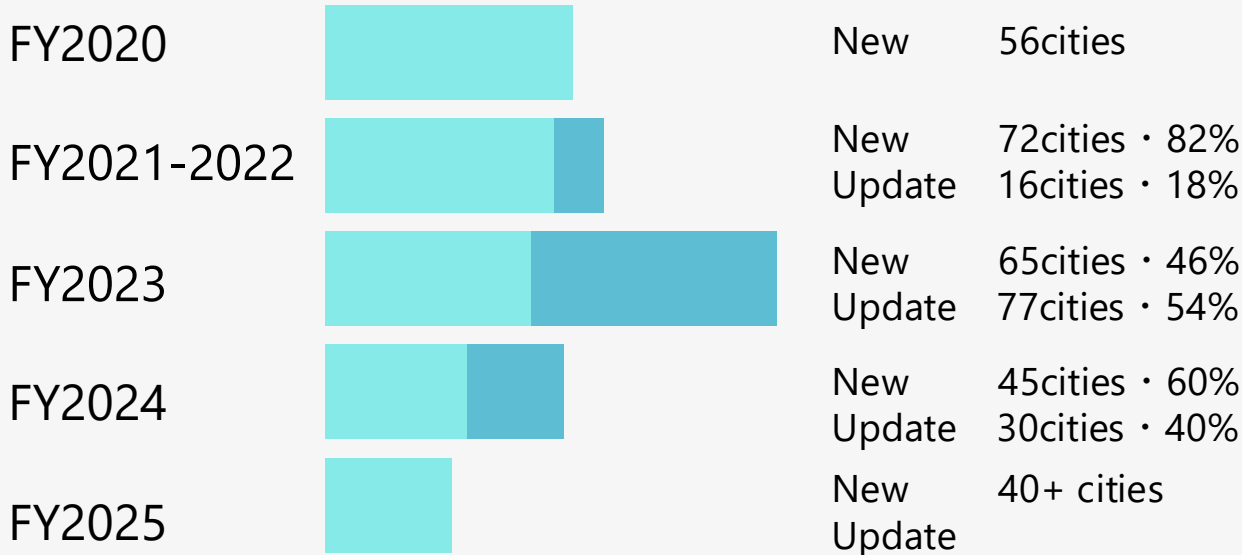
New Developed by



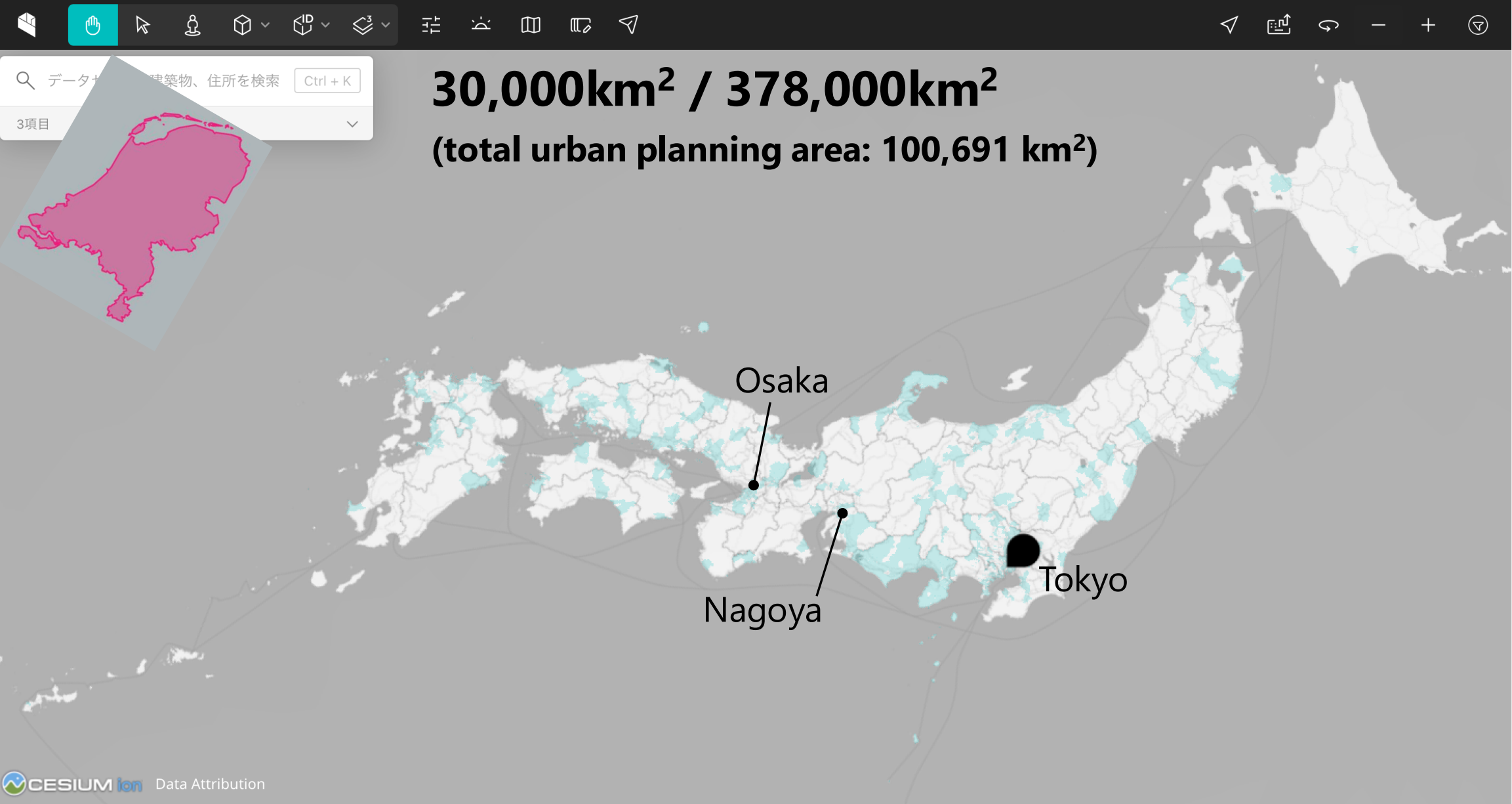
40+

FY2025

New development/Update



FY2020 - FY2025 Project PLATEAU Summary



PLATEAU VIEW

PLATEAU VIEW 5.0 <https://plateauview.mlit.go.jp/>



データセット、建築物、住所を検索 Ctrl + K

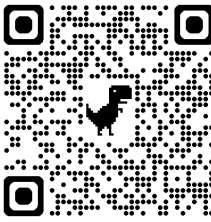
建築物モデル (千代田区)

建築物モデル (中央区)

全てを削除

2項目

東京都 / 千代田区

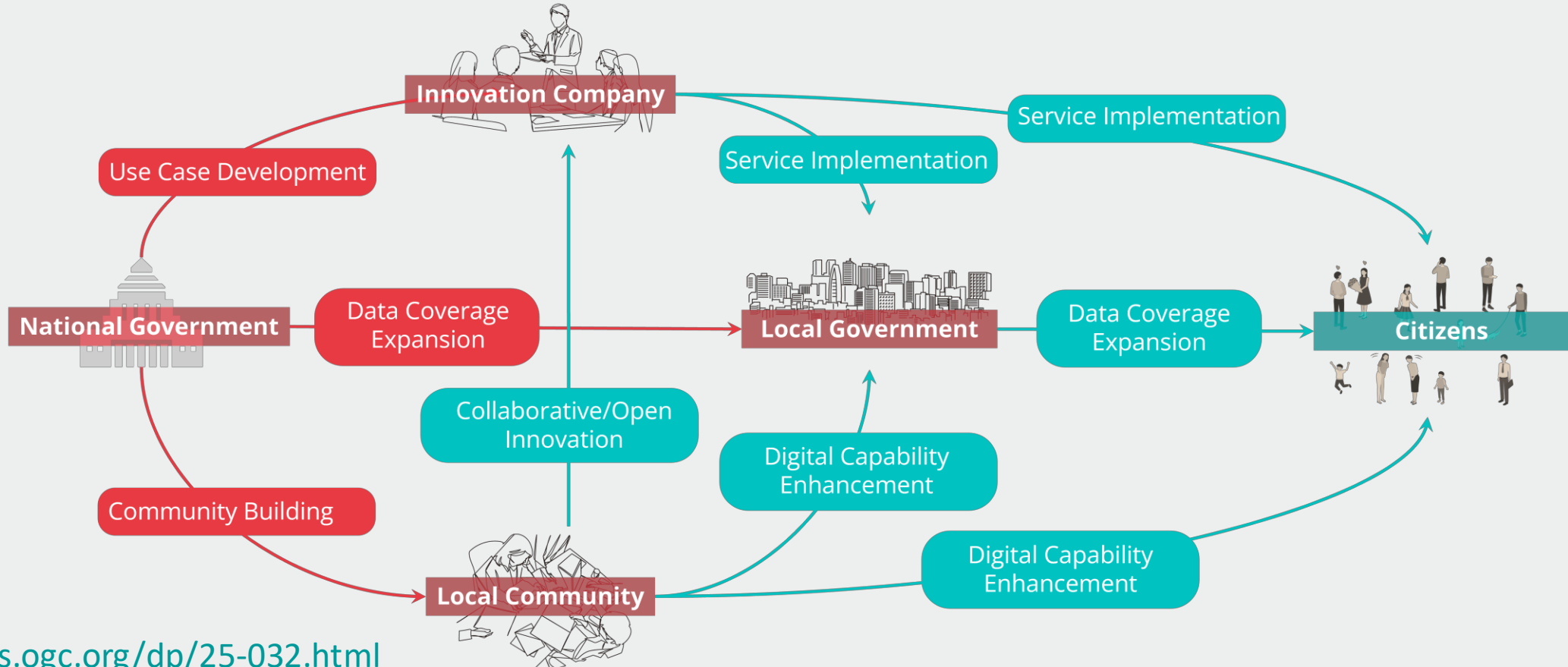


This video is rendered using LOD2-level building data and textures for the area around Tokyo Station.

Project Goal: Ecosystem

PLATEAU's goal is to create an ecosystem for the development, utilization, and open availability of 3D city models for more urban resilience.

- PLATEAU aims to build an 'ecosystem' in which the maintenance, utilization and open data conversion of 3D city models develop autonomously through the cooperation of diverse actors



PLATEAU Project Partners:

Over 300 local governments and 100 companies are joined



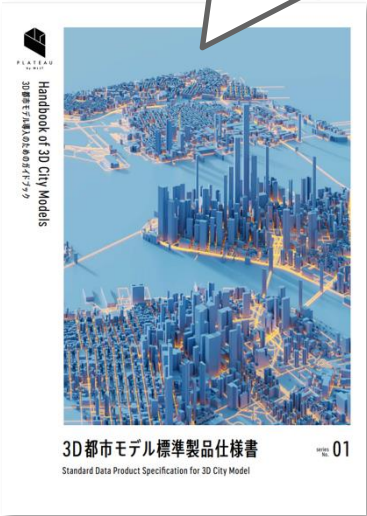
<https://www.mlit.go.jp/plateau/about/>

Development of Standard Data Product Specification +ADE for 3D city models

1,380 pages !

Standardization of 3D city model data products in Japan

- The Standard Data Product Specification for 3D City Model was developed in March 2021 as Japan's first standard data model for 3D city model, which was updated by **version 5.1 in March 2026**.
- The development of this standard document has resulted in the unification of the specifications, standards, and quality of 3D city models in Japan.
- As a result, software development will be more efficient, knowledge will be shared, and data will be linked more easily.



Compatible with international standards



- The PLATEAU standard data specification is based on **CityGML 2.0**, an open format developed by an international standards OGC.
- The PLATEAU standard is a localized standard unique to Japan that adds attribute information and detailed LOD definitions.

Migration to 3.0?



- HTML version was released in 2022 to improve usability

<https://www.mlit.go.jp/plateaudocument/>

Expanding the coverage of data

Establishment of high-quality and scalable data development

Urban Planning Base Map (1:1,500-2,500)

2D map data of
buildings, roads, blocks, etc.



Geometry (shape)

×

Aerial survey results

3D data of building **heights**,
shapes, **texture** etc.

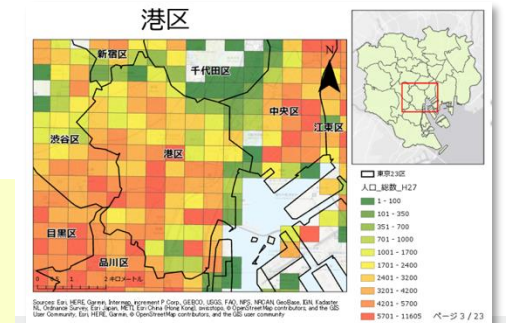


Buildings, Roads,
Bridges, Trees, etc.

3D City Models

Semantics(attribute)

Urban Planning Basic Survey data
Current status of each buildings and land, etc.



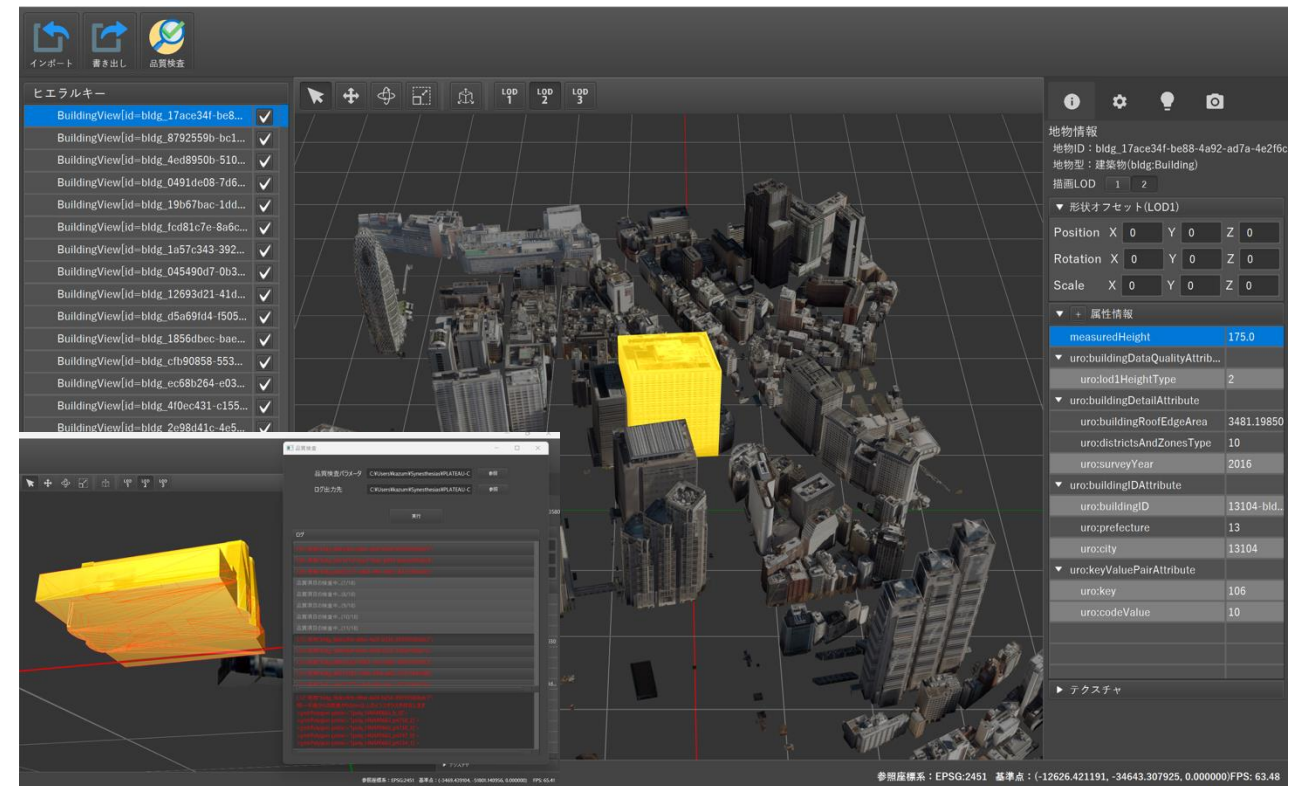
Data Quality Assurance in Project PLATEAU

Based on the “**Japanese Public Survey Act**” (since 1949)

Item	Requirement
Horizontal Accuracy (Map Information Level 1:2500)	Approx. 1.75 m
Vertical Accuracy	0.66 m
Topological Consistency	Required
Semantic Validation	Required
Metadata	Mandatory

QA Framework

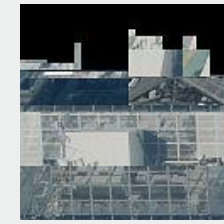
- Standard Product Specification
- Standard Workflow Specification
- Survey Manual
- Automated Validation Tools



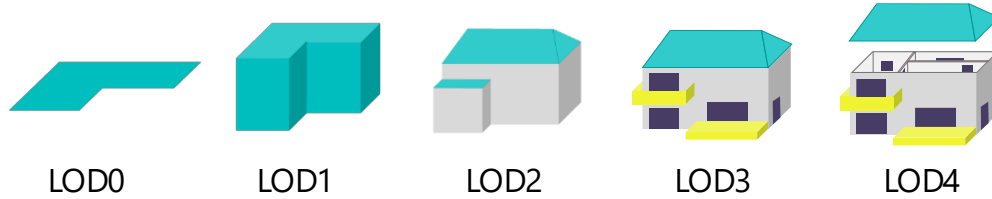
<https://github.com/Project-PLATEAU/PLATEAU-Builder>

Modules of the PLATEAU standard (examples)

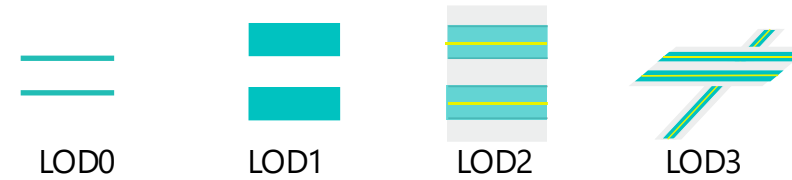
Textures are also included as separate files.



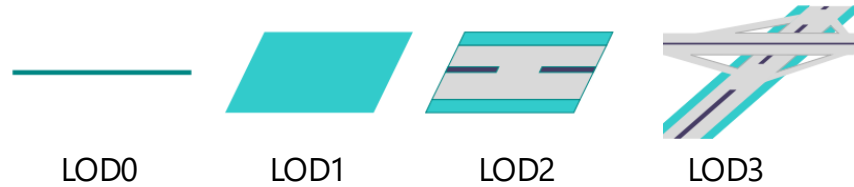
建築物モデル - Building



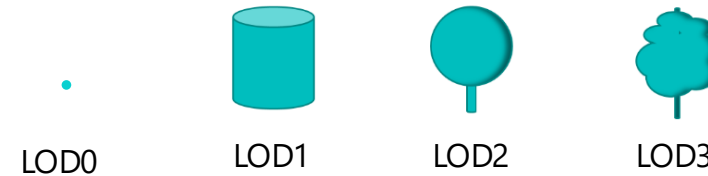
交通（鉄道）モデル - Transportation(Railway)



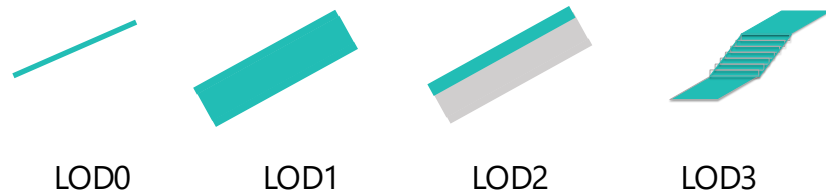
交通（道路）モデル - Transportation(Road)



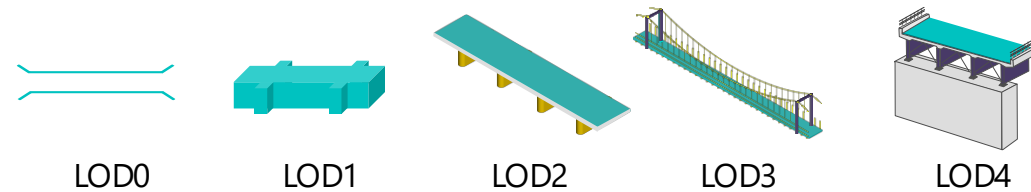
植生モデル - Vegetation



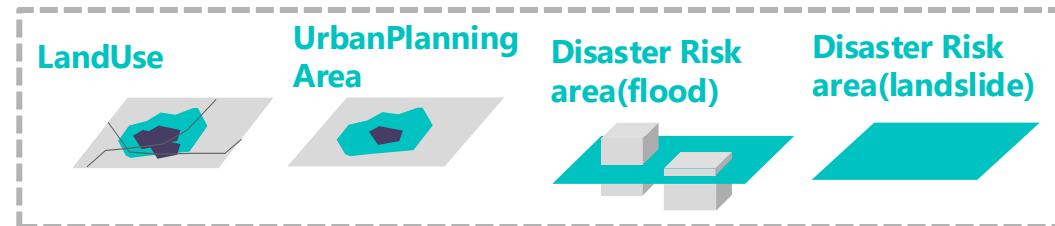
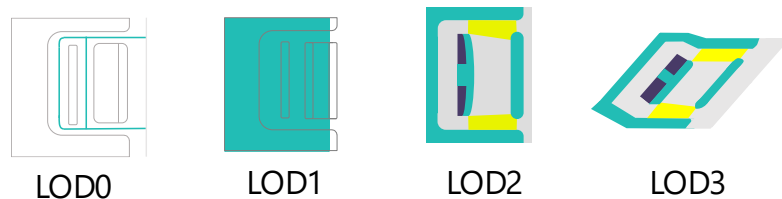
交通（徒歩道）モデル - Transportation(Track)



橋梁モデル - Bridge



交通（広場）モデル - Transportation(Square)



Why Does PLATEAU Use CityGML + ADE (Application Domain Extension) ?

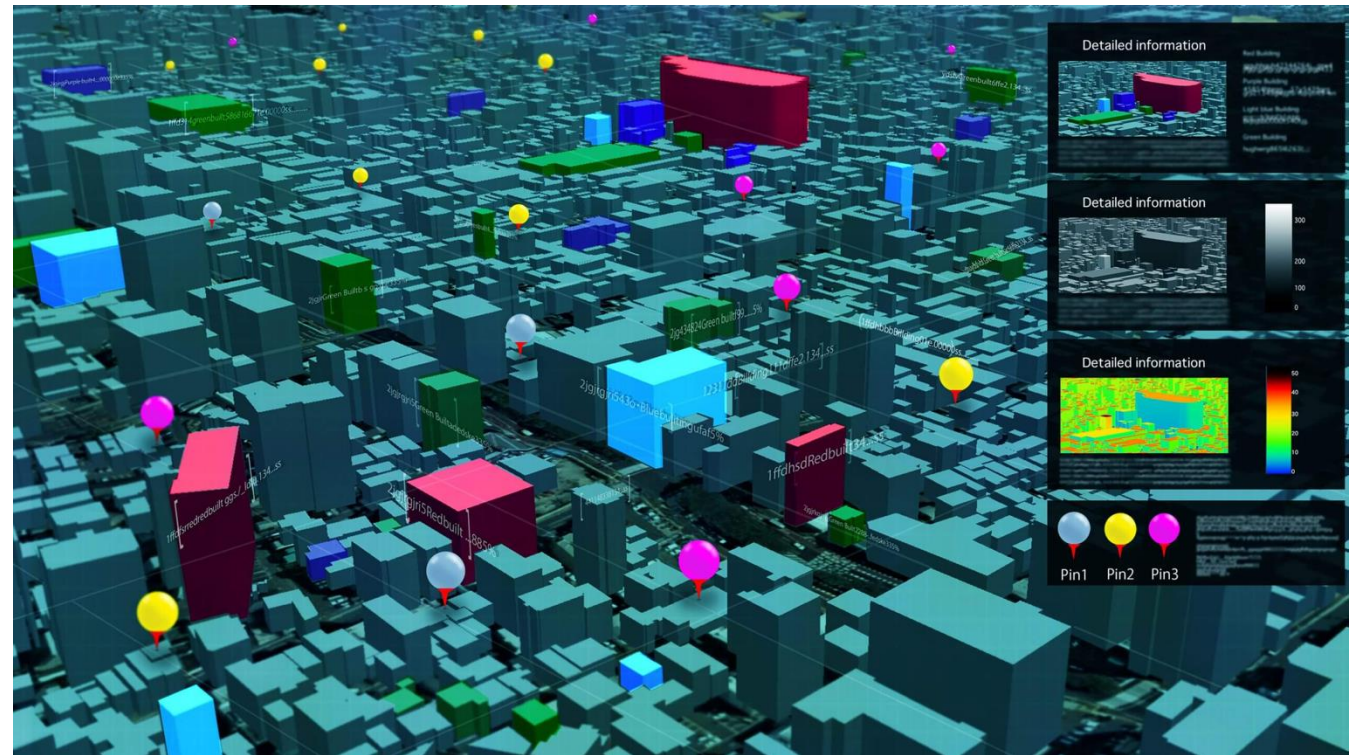
CityGML 2.0 alone does not provide sufficient urban planning semantics for Japanese planning systems.

- zoning
- land use regulations
- building coverage ratio
- urban redevelopment projects
- hazard risk zones



Japanese Urban Planning Requires

Physical City	Regulatory City
Buildings	Zoning
Roads	Planning restrictions
Parcels	Development regulations



<https://www.mlit.go.jp/plateau/use-case/uc22-042/>

PLATEAU View v5.0: Flood estimation area and each buildings forecast (color is rank)



Creating Best Practices for Use Cases

Creating digital twin-based solutions in diverse public and private sectors



Urban Planning

Use in planning and simulation for urban planning, development and area management, consensus building, and in apps for the community.



Disaster Management

Use in disaster prevention planning and simulation, risk visualization, evacuation route apps, disaster prevention workshops.



Revitalisation/Tourism

Used for creating metaverse spaces, XR tourism contents, tourism guide apps, advertising effectiveness simulations.



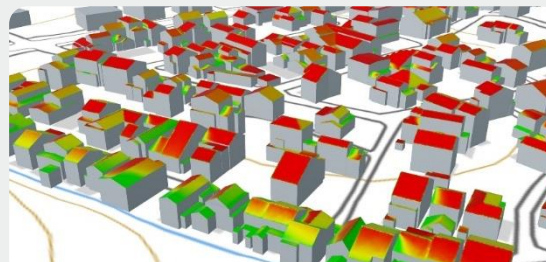
Mobility/Robots

Use for autonomous vehicles and drones for maps, OS and optimum route searches.



Public participation

Use for apps, XR tools, dashboards to support public participation in urban development and local activities



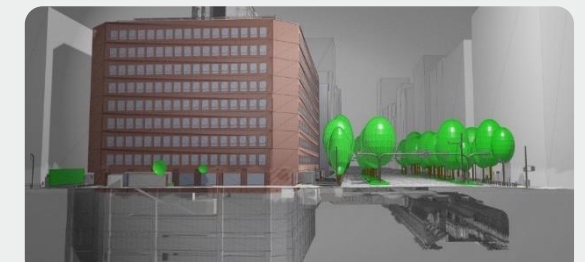
Environment/Energy

Use for simulation of solar power generation, heat islands, ventilation, and area energy management.



Infrastructure

Use for apps for buildings and park facilities management, ageing simulation and management through IoT data.



Digital twin tech

Development of technologies for segmentation and modelling of sensing data such as point clouds and integration with BIM.

- All use cases developed on PLATEAU are available on the website.
- Additionally, the technical insights gained during the process of use case development are published as technical reports.
- These materials are available to municipalities and others interested in similar use cases and applications.
- Sharing not only successful experiences but also failures as knowledge allows us to accelerate the social implementation of technology.

<https://www.mlit.go.jp/plateau/use-case/>

Use Case

3D都市モデルは社会課題の解決や新たな価値創造のポテンシャルを持っています
全国各地で実際に進められている、3D都市モデルを活用したソリューション開発の事例をご紹介します。

[See Use Case Movie](#)

Category **ALL** Field **Not selected** Tag **Not selected** 128 articles

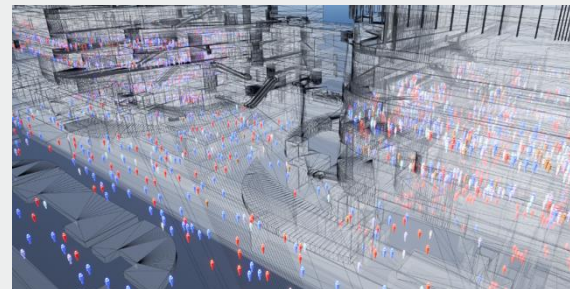
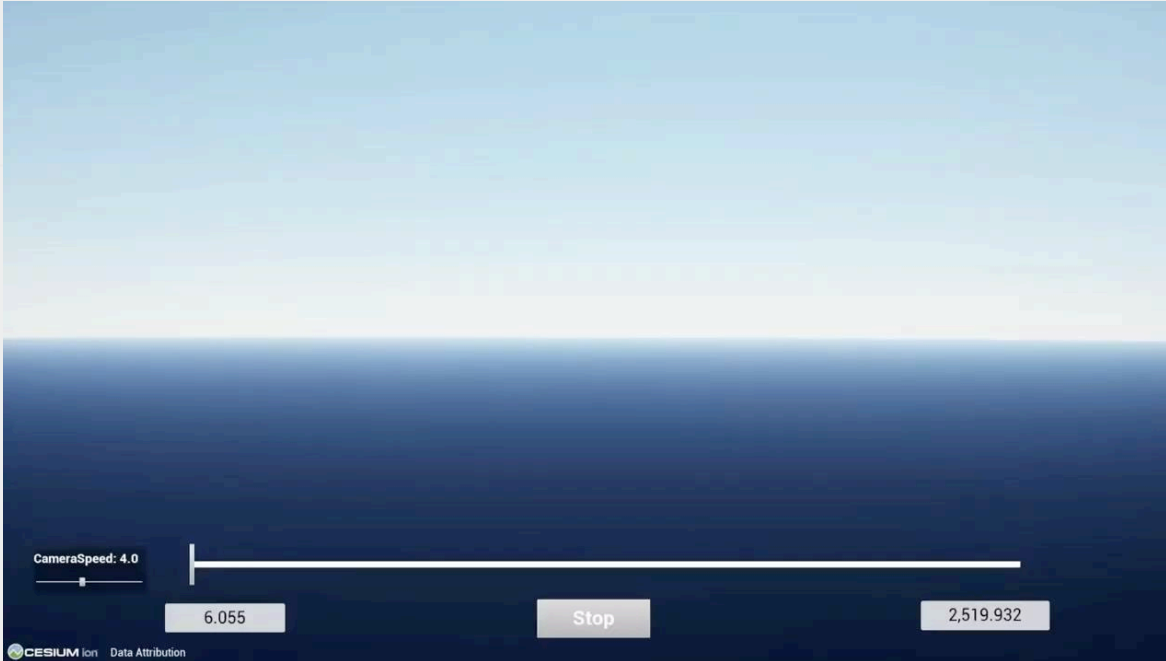
進行中プロジェクト 完了プロジェクト



Advanced Disaster Management in Area Design

Developed large-scale Crowd simulation using a 3D urban model to achieve DX in area design with a focus on disaster management.

- By integrating BIM, CAD, and 3D city models of the Shinagawa Station North District, we create an integrated indoor-outdoor model to facilitate large-scale guidance and evacuation simulations involving 10,000 people. This environment visualizes 3D representations of potential risks during disasters and the necessary evacuation plans to solve them.
- Leveraging these outcomes, we provide support for planning and consensus-building regarding evacuation in preparation for updating the Urban Regeneration Safety Assurance Plan (a disaster management plan in collaboration with the public and private sectors) for the area.



Agency: JR-EAST, KDDI CORPORATION, Tokyu Land Corporation, Nikken Sekkei Co., Ltd.

Location: Minato Ward, Tokyo



PLATEAU
by MLIT



Immersive urban planning with citizens using XR technology

Point

- Various ideas of participants can be visualized through the XR technology.
- The tool help citizens to join the urban development easily.



Tools for All

PLATEAU SDK tool kit for Unity

- A toolkit collection to support application development on Unity using the "3D city model" data provided by PLATEAU.
- Based on PLATEAU use cases, various types of applications using PLATEAU can be implemented with the required functionalities provided in four toolkits.



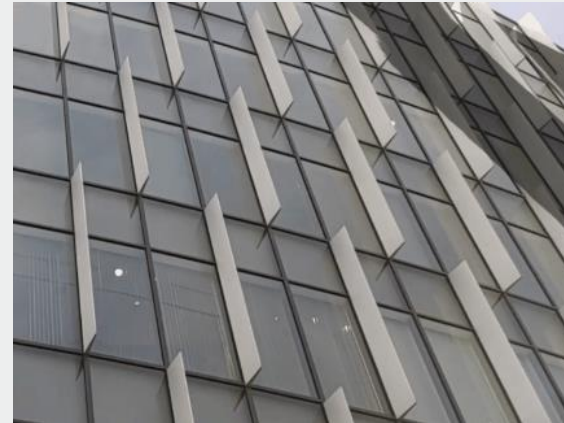
Render Toolkit

An environment system is provided that can change conditions such as location, time, and weather conditions.



Sandbox Toolkit

3D models of people, vehicles, and urban objects are provided as assets. Using a dedicated placement tool, automated movement settings can be configured.



AR Toolkit

Features are provided for utilizing 3D city models as occlusion objects in AR applications. Alignment functions using AR markers and Google's Geospatial API are also available.



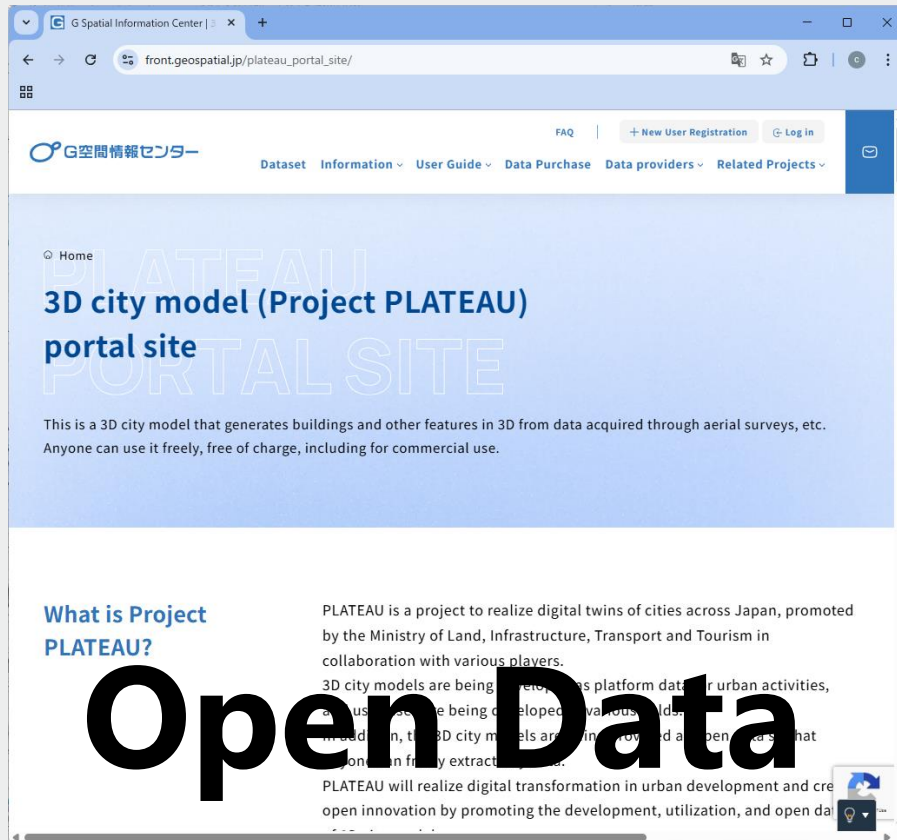
Maps Toolkit

A toolkit for spatial analysis, visualization, integration with architectural information, and the development of map applications using PLATEAU's 3D city models.

PLATEAU Achievements

Fostering Open Innovation

- PLATEAU not only shares the **3D city model as open data by CKAN** but also provides a variety of tools and other assets to utilize the 3D city model to foster open innovation.
- By adopting open licenses such as **CC BY 4.0**, we enable secondary use, modification, and commercial utilization, thereby promoting open innovation in various fields.



🔗 <https://www.geospatial.jp/ckan/dataset/plateau>



🔗 <https://github.com/Project-PLATEAU>

Searching PLATEAU Data with GenAI (experimental services)

 **PLATEAU 配信サービス** [List of datasets](#) [City GML](#) [3D Tiles / MVT](#) [Terrain](#) [Ortho](#) [API](#) [K](#) [🔄](#) [📄](#)

PLATEAU Delivery Service

PLATEAU 3D city model delivery service official document

[Quick Start →](#) [See the PLATEAU VIEW ↗](#) [GitHub ↗](#)

 **List of datasets** →
Filter search and copy the delivery URL

 **PLATEAU-CityGML** →
CityGML format data delivery specifications and CityGML API

 **PLATEAU-3DTILES / MVT** →
3D Tiles / MVT Data and Data Catalog API

 **PLATEAU-Terrain** →
Terrain model terraindb / Terrain-
RGB delivery

 **PLATEAU-Ortho** →
Delivery of aerial photo Orso image tiles

 **API Reference** →
REST API and GraphQL References & Playgrounds

 **MCP Server** →
Model Context Protocol Server for AI Clients

 **Agent Skills** →
Agent Skills for AI Coding Agents

 **Data conversion specification for visualization** →
CityGML to 3D Tiles/MVT Converter

PLATEAU's Initiatives

Fostering Open Innovation

Community building, enhancing digital capabilities for local governments, information dissemination, and developing support tools for developers.



Community Building

Encourage the formation of communities of engineers, citizens, and companies participating in PLATEAU through activities like app contests, hackathons, pitch events, hands-on workshops, and acceleration programs.

Enhancing Digital Capabilities

Strengthen digital capabilities in the public and private sectors to promote data utilization and support digital transformation by developing training programs for digital talent.

Creating Movement

Through information dissemination and media content creation, promote the usefulness of PLATEAU in various fields and stimulate a movement to encourage participation.

PLATEAU's Initiatives

Fostering Open Innovation : PLATEAU Next

PLATEAU Next, unique opportunities where you can experiment with new ideas and share the knowledge gained.

- ❑ Provide a platform to practice the "Creative Learning Spiral".
- ❑ Offer collaboration on 3D city models.
- ❑ Participant with diverse technical backgrounds, skill levels, specialties, and social positions.
- ❑ Promote cooperation and innovation.
- ❑ Facilitate the development of new talent.



PLATEAU Award !
(Competition events)

(in Japanese)

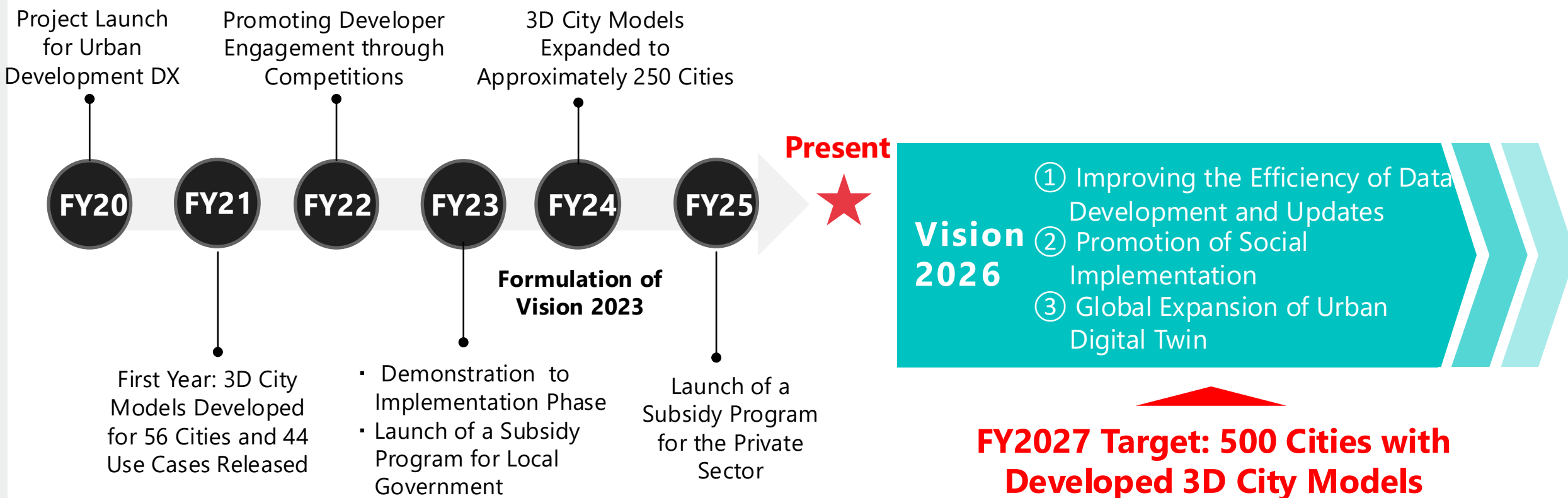


Lightning Talk, Hands-on and Hackathon events

7.1 (水) PLATEAU AWARD 2026 説明会 Online	Coming Soon PLATEAU Hands-on 22 Online On-site	8.8 (土) ~ 9 (日) PLATEAU × 関西イベント in 京都 (仮) Online On-site 学生対象	Coming Soon On-site 地域主催：沖縄												
Coming Soon PLATEAU LT 10 Online	Coming Soon PLATEAU Hands-on 23 Online On-site	8.22 (土) ~ 9.5 (土) PLATEAU CityHack Challenge 2026 Online On-site	Coming Soon On-site 地域主催：広島												
Coming Soon PLATEAU LT 11 Online	Coming Soon PLATEAU Hands-on 24 Online On-site	9.26 (土) ~ 27 (日) PLATEAU Hack Challenge 2026 in Tokyo On-site	Coming Soon On-site 地域主催：松江												
PLATEAU AWARD 2026			Coming Soon On-site 地域主催：八王子												
<table border="1"><tr><td>プレエントリー開始</td><td>2026年 6月22日(月)</td></tr><tr><td>プレエントリー募集締切</td><td>2026年 11月6日(金)</td></tr><tr><td>本エントリー募集開始</td><td>2026年 8月28日(金)</td></tr><tr><td>本エントリー募集締切</td><td>2026年 11月19日(木)</td></tr><tr><td>一次審査会</td><td>2026年 12月5日(土)~ 6日(日) Online</td></tr><tr><td>最終審査会</td><td>Coming Soon On-site</td></tr></table>			プレエントリー開始	2026年 6月22日(月)	プレエントリー募集締切	2026年 11月6日(金)	本エントリー募集開始	2026年 8月28日(金)	本エントリー募集締切	2026年 11月19日(木)	一次審査会	2026年 12月5日(土)~ 6日(日) Online	最終審査会	Coming Soon On-site	Coming Soon On-site 地域主催：京都
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最終審査会	Coming Soon On-site														

Progress on Initiatives

<Time schedule>



Challenges after Scaling PLATEAU

- ❑ Since its launch in 2020, Project PLATEAU has expanded the use of 3D city models by local governments and the private sector, **reaching 329 cities and around 100 use cases by FY2025.**
- ❑ However, high costs of data development and updates, quality management are key challenges. Especially for the **cost and gaps between user needs and data quality**, these would be the risks for data updates.
- ❑ To continue expanding the use of PLATEAU in urban policy and the creation of new markets, it is necessary to **establish new policy directions aimed at resolving these challenges.**

Key Challenges

- | | |
|--|--|
| ◆ Data Development and Updates Costs | * The high cost of model maintenance and updates is a burden on the entities responsible for maintenance. |
| ◆ Gaps between Needs and Data Quality | * Depending on the specific user needs, high-accuracy data may not always be necessary (Be flexible). |
| ◆ Social Implementation | * The integration of data utilization into daily works for urban development is still in progress. |
| ◆ Missing Opportunities | * Domestic activities limit the business opportunities and opportunities for technological innovation into Japan, resulting in lost opportunities in overseas markets. |

Three Key Priorities for the Future

1

Improving the Efficiency of Data Development and Updates

■ Expand data coverage and establish sustainable ecosystem (data development and updates) through cost-efficient methods.

- Promoting Needs-Based Data Development
- Development of Advanced Technologies for Efficient Data Development and Updates
- Leveraging Diverse Urban Data to Expand Urban Digital Twins

2

Promotion of Social Implementation

■ Establish the use of 3D City Models for daily works in urban development to address challenges in adjacent fields and expand across diverse sectors

- Advanced and Efficient Urban Planning & Administrative Works
- Fostering Business Innovation to Address Social Challenges
- Greater Value Creation for Diverse Use

3

Global Expansion of Urban Digital Twin

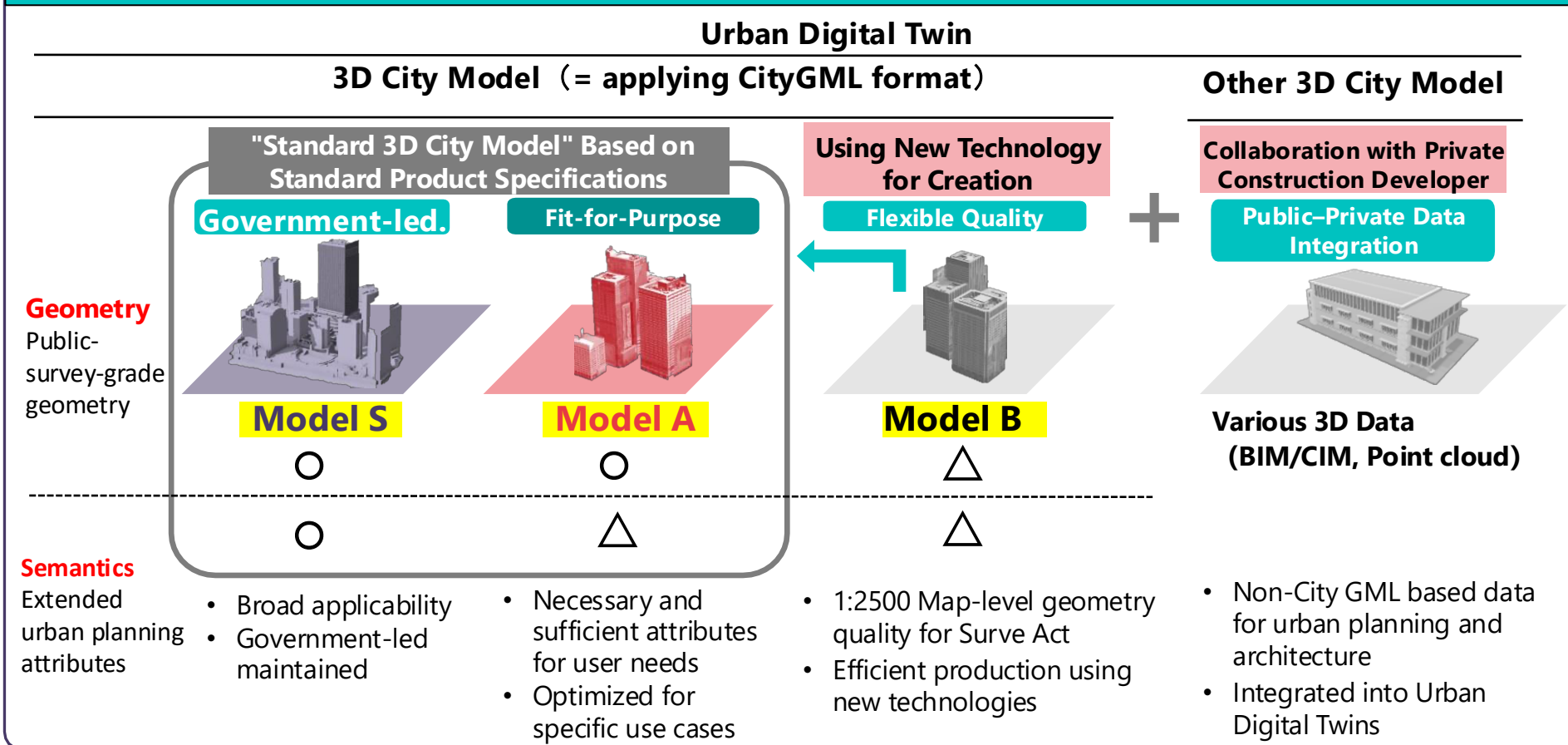
■ Promoting Global Expansion of Urban Digital Twin and Leveraging Feedbacks for Domestic Initiatives

- Enhancing the International Reputation and Mainstreaming of Japan's Initiatives
- Alignment with International Urban Development Initiatives

Key Priorities for the Future: (i) Efficient Data Creation and Updates

- Expand beyond CityGML-based 3D city models
- Support multiple data sources and quality levels
- Reduce update costs while increasing usability

① Expand the Data Variety consists of Project PLATEAU



② Social Implementation

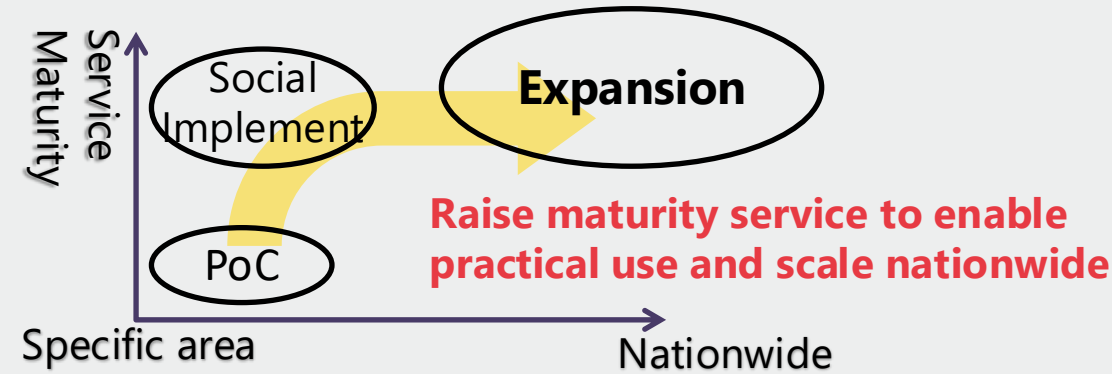
- Administrative DX
- New business creation
- Greater user value

③ Global Expansion

- Diverse urban environments
- Scalable data production

Key Priorities for the Future: (ii) Further Promotion of Social Implementation

- ❑ Move from PoC to daily-routine operations
- ❑ Integrate digital twins into urban planning and administration
- ❑ Scale successful services nationwide



Urban
planning
administr
ation

○ Upgrading the Administrative Processes related to Urban Planning, including Operations, Procedures and Management.

- Use 3D City Model in daily administrative works and operate sustainably, promoting DX.
- Promote the simulation and EBPM in urban planning as well as disaster risk reduction.
- Integrated data management for efficient urban infrastructure management including construction and inspection.
- Phase-free utilization in disaster management at the stage of emergency response, and recovery and reconstruction planning.

Fields
related to
Urban
Planning

○ Fostering Business Innovation to Tackle Social Challenges

- Planning, design, and development for environmental-friendly and sustainably innovation.
- Promote the creation of new services that address social challenges.
- Expansion of business opportunities and value creation that contribute to tackle social challenges.

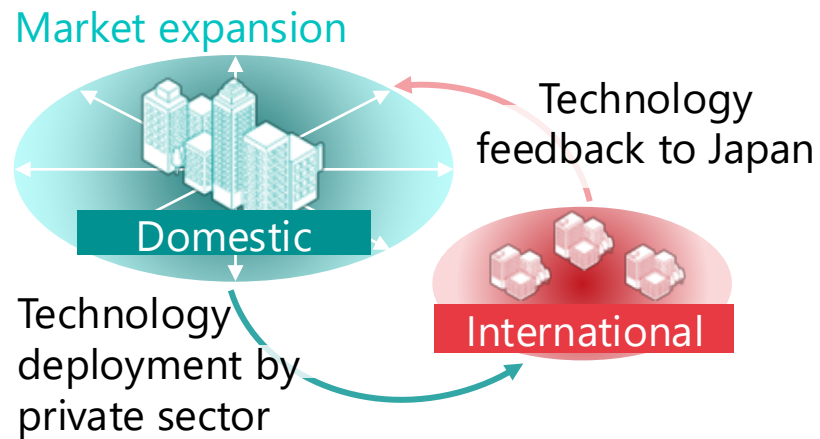
Other
fields

○ Further Value Creation to Expand Users

- Increase usability through broader data coverage.
- Explore the value of 3D City Model by expand users.

Key Priorities for the Future: (iii) Global Expansion of Urban Digital Twins

- ❑ Growing global demand for Urban Digital Twins
- ❑ Promote Japanese UDT initiatives internationally
- ❑ Leverage international experience for domestic growth



Flood Disaster Prediction in Ghana
Source: Space Data Inc.



Pedestrian Deck Planning in Dukuh Atas,
Indonesia Source: MLIT

【Branding and Mainstreaming】

- Promote PLATEAU through international organizations and partnerships
- Contribute to OGC standardization activities

【Linking with Urban Development Initiatives】

- Expand collaboration with governments, academia, and industry
- Integrate UDT solutions into urban development projects.

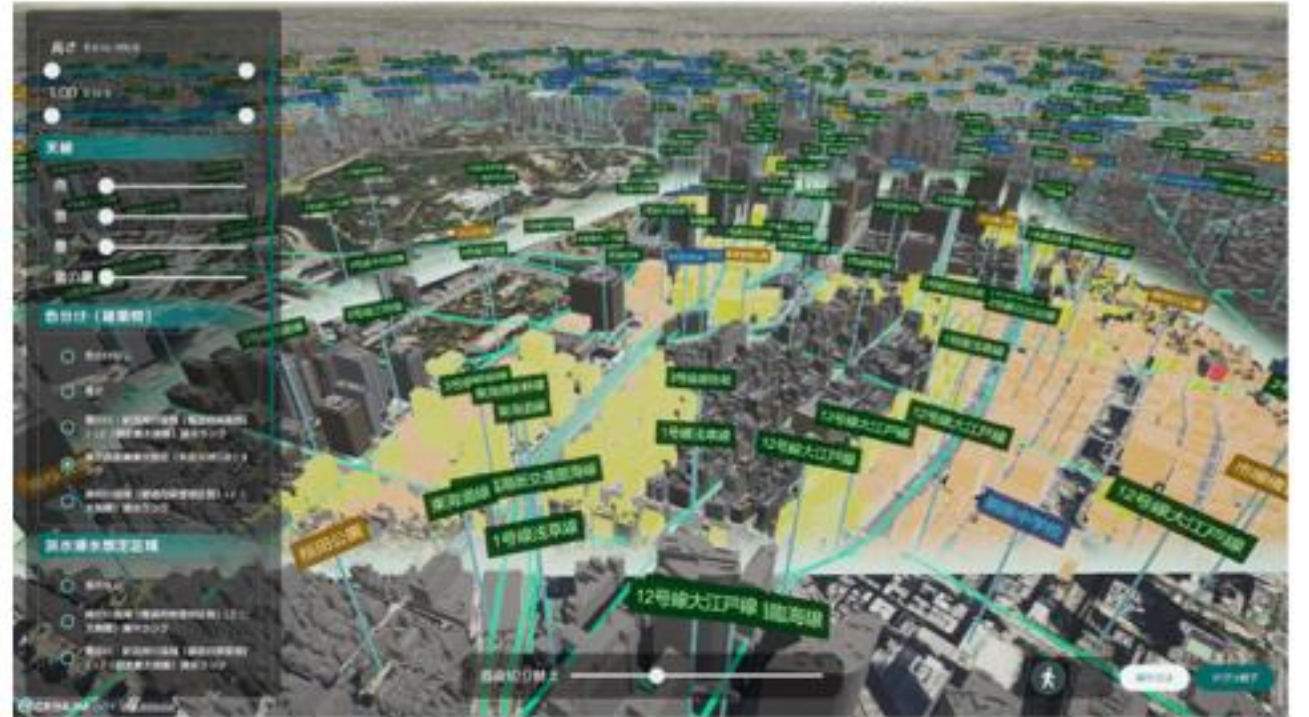
Key Challenge: Quality and/or Sustainability

Current PLATEAU

- Public survey grade quality
- Nationwide standardization
- Rich semantics

However,

- High update costs
- Long production cycles
- Municipal budget constraints



Key Question

What level of quality is sufficient for sustainable maintenance?

This is one of the most important governance challenges facing PLATEAU today.

Future Topics for Japan-Netherlands Collaboration

- **Standardization**
 - File formats (CityJSON, OGC 3D Tiles, CityGML 3.0 etc..)
 - GeoBIM Integration
- **Data Supply**
 - OGC API-based data delivery
 - Cloud-native SDI
- **Quality Management**
 - Fit-for-purpose datasets
- **Governance**
 - Sustainable update models



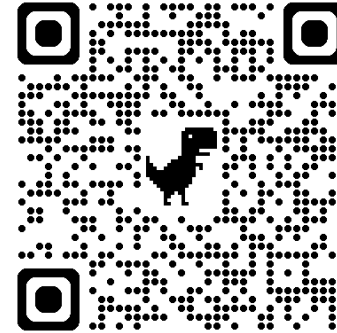
PLATEAU
by MLIT

Dank u wel / Thank You !

PLATEAU WEB



PLATEAU View



PLATEAU GitHub



PLATEAU API/MCP
tools (experimental)

